MILITARY SPECIFICATION

RING LUCKY AND INSULATOR ASSEMBLY FOR CARTRIDGE, 105MM, HEAT-T, M456A1E1 PARTS FOR, ASSEMBLY AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, assembly, and packing for one type of Ring Lucky and Insulator Assembly for Cartridge, 105MM, HEAT-T, M456A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1315

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DRAWINGS

ARMY

9256836 - Ring Lucky and Insulator Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Ring Lucky and Insulator Assembly.-The ring lucky and insulator assembly shall comply with all requirements specified on drawing (dwg.) 9256836, and on all other applicable drawings and specifications.

3.2.1 Mechanical Strength.-The segment lucky shall withstand a compressive load of 50,000 psi minimum (min.) without cracking or yielding.

3.2.2 Lucky Activity and Polarity.-The segment lucky shall have a piezo-electric coefficient, d_{33}, of not less than 15×10^{-10} coulombs per pound where d_{33} relates the charge to the load and is measured along the axis of polarization. Polarity markings shall be as specified.

3.2.3 The dielectric constant of the segment lucky shall not vary more than plus or minus ($\pm$) 40 percent (%) from its room temperature value as its temperature is continuously varied over the range of -65 degrees Fahrenheit (°F) to 160 °F.

3.2.4 Capacitance.-The segment lucky shall not have a capacitance of less than 1400×10^{-12} farads nor more than 2000×10^{-12} farads.

3.2.5 Ring Lucky Assembly Activity and Polarity.-Each segment lucky on the ring electrode shall have a piezo-electric coefficient, d_{33}, of not less than 9×10^{-10} coulombs per pound where d_{33} relates the charge to the load and is measured

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along the axis of polarization. Activity shall indicate that the positive poled lucky face is cemented to the ring electrode.

3.2.6 Lucky Assembly Voltage Breakdown.-Each segment lucky on the ring lucky assembly shall withstand application of 1500 volts min. direct current (DC) for a min. duration of 2 seconds, without visible disruptive discharge.

3.2.7 Lead Wire Pull Test.-The lead wire joint must withstand a direct pull of forty (40) pounds, minimum.

3.2.8 Adhesion.-Adhesive bonded parts of the ring lucky assembly shall not become loose or unbonded after being subjected to the adhesion test.

3.2.9 Insulator Voltage Breakdown.-The insulator shall withstand minimum application of 2000 volts DC for a duration of 2 seconds, minimum without visible disruptive discharge.

3.2.10 Insulator Resistance.-Electrical resistance measured across the rear web of the insulator shall be no less than 10^6 ohms.

3.3 Workmanship.-The assembly and all components shall be of the proper finish, free of cracks or burrs; and chips, dirt, grease, rust or other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.4 First Article.-This specification makes provisions for submission of first article sample inspection in 4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109

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to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

f. Provisions for submission of first article samples.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to initiation of regular production, the contractor shall submit a first article sample to the Government facility designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) assemblies and twenty-five (25) complete sets of parts which have been produced by the contractor or furnished by a supplier, and which

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have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply so as to significantly affect product uniformity as determined by the Government.)

4.2.2 Inspection to be Performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2, 4.3.3 and 4.3.4 of this specification, and any or all of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements for the inspections performed in accordance with 4.3.2, 4.3.3 and 4.3.4, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly or component in the sample to comply with any or all of the stated requirements.

4.3 Quality Conformance Inspection

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification

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revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.1.1 Curing Batch (See paragraph 4.3.4.10).-A batch shall be defined as the number of assemblies cured under the same conditions, in the same oven, at one time.

4.3.2 Inspection Molded Parts - Insulator.-A sample of ten (10) ring insulators shall be randomly selected from each hour's production and subjected to the inspections as specified in 4.3.3.5. In addition, whenever production is resumed after a downtime period or at the start of a new shift, the first five (5) insulators produced shall also be inspected. If one or more insulators is found to be defective, the group represented by the sample and the last accepted batch shall be screened 100%.

4.3.3 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.5.

4.3.3.1 Segment Electrode (See dwg. 9256833).

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Categories	Defects	Method of Inspection (See 6.3)
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 1.50 Percent	
201.	Flatness incorrect	Gage
202.	Knurl or stipple missing or incomplete	Visual
203.	Profile incorrect	Gage
204.	Evidence of poor workmanship (See 3.3)	Visual

4.3.3.2 Ring Electrode (See dwg. 9256831).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 2.50 Percent	
201.	Distance from bottom of slot to outside diameter incorrect	Gage
202.	Symmetry of hole with ring faces incorrect	Gage
203.	Width of slot incorrect	Gage
204.	Inside diameter incorrect	Gage
205.	Knurl or stipple missing or incomplete	Visual
206.	Outside diameter incorrect	Gage
207.	Evidence of poor workmanship	Visual

4.3.3.3 Segment Lucky (See dwg. 9256832).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 Percent	
101.	Profile incorrect	Gage
102.	Flatness incorrect	Gage

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103. Coating improper or missing.....Visual
104. Polarity mark improper or missing...Visual

Minor: AQL 2.50 Percent
201. Uncoated borders incorrect.....Gage
202. Faces of segment lucky not parallel..Gage
203. Conductive paint or particle on areas other than designated.....Visual
204. Thickness incorrect.....Gage
205. Evidence of poor workmanship.....Visual

4.3.3.4 Ring Lucky Assembly (See dwg. 9256835).

4.3.3.4.1 Prior to Cementing Components.

4.3.3.4.1.1 Before Soldering Wire.

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.40 Percent
101. Wire strands nicked.....Visual

Minor: AQL 1.50 Percent
201. Joint fails to withstand 5-pound (lb.) pull test.....Test
202. Length of hook under min.....Gage
203. Length from stake to inside diameter of ring electrode under min.....Gage

4.3.3.4.1.2 After Soldering Wire

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.65 Percent
101. Joint protrudes above surfaces of ring electrode.....Gage
102. Flux not removed.....Visual
103. Evidence of joint corrosion.....Visual
104. Flatness incorrect.....Gage

Minor: AQL 0.65 Percent
201. Joint protrudes beyond outside diameter of ring electrode.....Gage

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4.3.3.4.1.3 After Cementing Luckies.

Categories	Defects	Method of Inspection
Critical:	None Defined.	
Major:	AQL 0.40 Percent	
101.	Lucky assembled to electrode improperly	Visual
Minor:	None defined.	

4.3.3.4.2 After Cementing and Curing.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 Percent	
101.	Length of wire incorrect	Gage
102.	Insulation cracked, embritted, burnt or damaged	Manual/Visual
103.	Wire strands nicked or broken	Visual
104.	Luckies protrude beyond electrode ring greater than permitted	Manual/Visual
105.	Lucky spacing incorrect	Gage
106.	Segment electrodes overhand luckies	Visual/Manual
Minor:	AQL 2.50 Percent	
201.	Electrodes too close to edge of lucky at ends	Gage
202.	Epoxy on edges of lucky	Visual
203.	Flatness requirement incorrect	Gage
204.	Thickness over max	Gage
205.	Evidence of poor workmanship	Visual

4.3.3.5 Ring Insulator (See 4.3.2 and dwg. 9256834). (Inspections to be performed while item is in restrained position).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Break in rubber	Visual

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Minor:
201. Outside diameter incorrect.....Gage
202. Inside diameter incorrect.....Gage
203. Thickness incorrect.....Gage
204. Length of nozzle incorrect.....Gage
205. Evidence of flash.....Visual

4.3.3.6 Ring Lucky and Insulator Assembly (See dwg. 9256836).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	AQL 0.65 Percent	
101.	Lucky segments not visible.....	Visual
102.	Components loose or missing.....	Visual/Manual
103.	Insulator cut or damaged.....	Visual
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship.....	Visual

4.3.4 Testing

4.3.4.1 Mechanical Strength (See 3.2.1) - Major Defect Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I. If one or more samples fail to comply with the applicable requirement, the lot shall be rejected. The samples subjected to this test shall not be returned to the lot. Testing shall be performed using contractor designed equipment in accordance with 4.4.1.

4.3.4.2 Lucky Activity (See 3.2.2) - Major Defect - and Polarity (See dwg. 9256835) - Major Defect. This test shall be performed 100%. Any lucky segment that fails to comply with the applicable requirement shall be removed from the lot. Testing shall be performed using contractor designed equipment in accordance with 4.4.2.

4.3.4.3 High Force Lucky Activity (See 3.2.2) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I. If one or more samples fail to comply with the applicable requirement, the lot shall be rejected. The samples subjected to this test shall not be returned to the lot. Testing shall be performed using contractor designed equipment in accordance with 4.4.3.

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4.3.4.4 Temperature-Aged Lucky Activity (See 3.2.2) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I, AQL 0.40%. Acceptable test samples may be returned to the lot after the residual charges have been removed. Testing shall be performed using contractor designed equipment in accordance with 4.4.4.

4.3.4.5 Dielectric Constant (See 3.2.3) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I. If one or more samples fail to comply with the applicable requirement, the lot shall be rejected. Samples may be returned to the lot after re-testing for compliance with 3.2.2, using the procedure of 4.2.2. Testing shall be performed using contractor designed equipment in accordance with 4.4.5.

4.3.4.6 Capacitance (See 3.2.4) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level II, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.6.

4.3.4.7 Ring Lucky Assembly Activity (See 3.2.5) - Major Defect and Polarity (See 3.2.5) Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level II, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.7.

4.3.4.8 Lucky Assembly Voltage Breakdown (See 3.2.6) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level II, AQL 0.40%. Acceptable test samples may be returned to the lot after the residual charges have been removed and assemblies have been retested for compliance with 3.2.5. Testing shall be performed using contractor designed equipment in accordance with 4.4.8.

4.3.4.9 Lead Wire Pull Test (See 3.2.7) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I, AQL 0.40%. Samples subjected to this test shall be the lowest level assembly of components which contains the finished soldered wire and ring electrode joint. Testing shall be performed using contractor designed equipment in accordance with 4.4.9.

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4.3.4.10 Adhesion (See 3.2.8) - Major Defect - Samples selected from each curing batch (See 4.3.1.1) shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.10.

4.3.4.11 Insulator Voltage Breakdown (See 3.2.9) - Major Defect - Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level II, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.11.

4.3.4.12 Insulator Resistance (See 3.2.10) - Major Defect - -Samples shall be submitted and tested in accordance with MIL-STD-105, General Inspection Level I, AQL 0.40%. Testing shall be performed using contractor designed equipment in accordance with 4.4.12.

4.3.5 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.3.3 and in 4.3.4 (See 6.2).

4.4 Test Methods and Procedures

4.4.1 Mechanical Strength.-The required load shall be applied across the lucky faces and retained for a minimum duration of 30 seconds. The test equipment shall be in accordance with 4.3.5.

4.4.2 Lucky Activity and Polarity.-The piezo-electric coefficient shall be determined by measuring the charge developed while releasing the lucky from a compressive pressure of 1000-50 pounds per square inch (psi) of electrode area (1 side). The load shall be distributed across the electroded areas by anvils whose faces are flat and parallel to each other. The following procedure shall be used:

a. Load sample to 1000-50 psi with fixture output leads shorted.

b. Remove short from sample and connect leads to an electrometer with input impedence of 10^{12} ohms min. and capacitance of 50×10^{-12} farads max. The input is to be swamped with a capacitor of 2×10^{-6} farads min. Instrument readout shall indicate polarity.

c. Release load; measure voltage output.

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d. Determine d_{33} (piezo-electric coefficient) from the equation:

$$d_{33} = \frac{CV}{F}$$

Where:

C = swamping capacitance in farads
V = voltage output in volts
F = applied force in pounds

Test equipment shall be in accordance with 4.3.5.

4.4.3 High Force Lucky Activity.-The piezo-electric coefficient (d_{33}) for the segment lucky shall be determined as specified in 4.4.2, but using a compressive load of 10,000 - 500 psi. Test equipment shall be in accordance with 4.3.5.

4.4.4 Temperature-Aged Lucky Activity.-The piezo-electric coefficient (d_{33}) for the segment lucky shall be determined as specified in 4.4.2, after the samples have been subjected to the following temperature conditioning:

a. Raise temperature to 160 ° - 5°F and hold for one hour min.

b. Lower to room temperature and hold for one hour min.

c. Lower temperature to -60 ° - 5°F and hold for one hour min.

d. Raise to room temperature and hold for 48 hours min.

Test equipment shall be in accordance with 4.3.5.

4.4.5 Dielectric Constant.-The dielectric constant shall be continuously recorded versus temperature using standard instrumentation Temperature control shall be within ± 10°F. Allow a minimum of 10 minutes soak time at the temperature extremes. Test equipment shall be in accordance with 4.3.5.

4.4.6 Capacitance.-Capacitance of the segment lucky shall be measured at 1000 Hertz using any standard capacitance meter. Test equipment shall be in accordance with 4.3.5.

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4.4.7 Ring Lucky Assembly Activity and Polarity.-d_{33} and polarity of the individual segment luckies shall be determined as specified in 4.4.2. Test equipment shall be in accordance with 4.3.5.

4.4.8 Lucky Assembly Voltage Breakdown.-The test voltage shall be applied across the luckies with the positive side connected to the ring electrode. The test may be performed with luckies in parallel or on each individual lucky. The voltage source may be any standard DC power source with an output impedence of 5000 ohms max. Test equipment shall be in accordance with 4.3.5.

4.4.9 Lead Wire Pull Test.-The integrity of the soldered joint shall be tested by applying a tensile load of 40 lbs. min. between the wire and the section of the inside diameter of the ring electrode which surrounds the joint hole. The load shall be applied for a duration of 5 seconds min., and shall provide a pull which is in line with the hole axis. Test equipment shall be in accordance with 4.3.5.

4.4.10 Adhesion.-The adhesive bonds between the ring electrode and each segment lucky shall be loaded in a shear stress by application of a 20-1 lb. load. Bonds between segment electrodes and luckies shall be tested manually. Test equipment shall be in accordance with 4.3.5.

4.4.11 Insulator Voltage Breakdown.-The insulator shall be subjected to 2000 volts DC minimum, applied to the inside and outside faces. The test shall be performed in accordance with Method 301 of MIL-STD-202, with visual observation for voltage breakdown (See 6.4). The equipment for this test shall be in accordance with 4.3.5.

4.4.12 Insulator Resistance.-The electrical resistance measurement shall be taken across the entire rear face of the insulator and the entire inside surface of the rear web while the web is compressed by a force of 80 lbs. min. The min. test voltage shall be 500 volts D.C. The test fixture shall be appropriately shielded for safety. Test equipment shall be in accordance with 4.3.5.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

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5.1.1 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions of submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified

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equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification, paragraph and defect code numbers (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-A, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 It is noted that the size of inspection lots of component assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Voltage Breakdown.-Voltage breakdown is a visible disruptive discharge as evidenced by flashover (surface discharge) or sparkover (air discharge) between any segment electrode and the ring electrode or between adjacent segment electrodes or on the insulator surface.

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6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, when the assembly is procured by the Army Armaments Command, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A and
SARPA-AD-E-A
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1315-A-010

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